WORKED SOLUTION

STEP II 1988, Question 6

Pure • Curve sketching

STEP II 1988, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s2-q6

The key idea

Each of the three functions is zero at $x = 0$, so it is enough to show that it is increasing for $x > 0$. The derivatives are chosen so that each part feeds the next: the derivative of (ii) contains the function from (i), and the derivative of (iii) is a rescaled copy of (ii). For f, logarithmic differentiation turns the sign of f′ into the sign of the function in (iii), which is clearly what that part was set up for.

Solution

Throughout, we use the principle that if $F(0) = 0$ and $F'(x) > 0$ for all $x > 0$, then $F(x) > 0$ for all $x > 0$. (By the mean value theorem, $F(x) = F(x) - F(0) = xF'(\xi)$ for some $\xi \in (0, x)$.)

Part (i)

Let $g(x) = x - \tanh x$. Then $g(0) = 0$ and

$$g'(x) = 1 - \operatorname{sech}^2 x = \tanh^2 x,$$

using $1 - \tanh^2 x = \operatorname{sech}^2 x$. For $x > 0$ we have $\tanh x > 0$, so $g'(x) > 0$. Hence $x - \tanh x > 0$ for $x > 0$.

Part (ii)

Let $h(x) = x \sinh x - 2 \cosh x + 2$. Then $h(0) = 0 - 2 + 2 = 0$ and

$$h'(x) = \sinh x + x \cosh x - 2 \sinh x = x \cosh x - \sinh x = \cosh x \, (x - \tanh x).$$

For $x > 0$, $\cosh x > 0$ and $x - \tanh x > 0$ by part (i), so $h'(x) > 0$. Hence $h(x) > 0$ for $x > 0$.

Part (iii)

Let $k(x) = 2x\cosh 2x - 3\sinh 2x + 4x$. Then $k(0) = 0$ and

$$\begin{aligned} k'(x) &= 2\cosh 2x + 4x\sinh 2x - 6\cosh 2x + 4 \\ &= 4x\sinh 2x - 4\cosh 2x + 4. \end{aligned}$$

? **Ask yourself** Where have we seen this combination before?

Writing $u = 2x$, this is $2u\sinh u - 4\cosh u + 4 = 2\left(u\sinh u - 2\cosh u + 2\right) = 2h(u)$. So

$$k'(x) = 2h(2x) > 0 \quad \text{for } x > 0,$$

by part (ii), since $2x > 0$. Hence $k(x) > 0$ for $x > 0$.

No turning points of f

For $x > 0$ every factor in $\mathrm{f}(x)$ is positive, so $\mathrm{f}(x) > 0$ and we may take logarithms:

$$\ln \mathrm{f}(x) = \ln x + \tfrac{1}{3}\ln\cosh x - \ln\sinh x.$$

Differentiating,

$$\frac{\mathrm{f}'(x)}{\mathrm{f}(x)} = \frac{1}{x} + \frac{\sinh x}{3\cosh x} - \frac{\cosh x}{\sinh x} = \frac{3\sinh x\cosh x + x\sinh^2 x - 3x\cosh^2 x}{3x\sinh x\cosh x}.$$

Now we put the numerator in terms of double angles, using $\sinh x\cosh x = \frac{1}{2}\sinh 2x$, $\sinh^2 x = \frac{1}{2}(\cosh 2x - 1)$ and $\cosh^2 x = \frac{1}{2}(\cosh 2x + 1)$:

$$\begin{aligned} 3\sinh x\cosh x + x\sinh^2 x - 3x\cosh^2 x &= \tfrac{3}{2}\sinh 2x + \tfrac{x}{2}(\cosh 2x - 1) - \tfrac{3x}{2}(\cosh 2x + 1) \\ &= \tfrac{3}{2}\sinh 2x - x\cosh 2x - 2x \\ &= -\tfrac{1}{2}(2x\cosh 2x - 3\sinh 2x + 4x). \end{aligned}$$

Therefore

$$\mathrm{f}'(x) = -\frac{\mathrm{f}(x)\left(2x\cosh 2x - 3\sinh 2x + 4x\right)}{6x\sinh x\cosh x}.$$

For $x > 0$, the bracket is positive by part (iii), and $\mathrm{f}(x)$, x, $\sinh x$ and $\cosh x$ are all positive. So $\mathrm{f}'(x) < 0$ for every $x > 0$: f′ never vanishes, so f has no turning points, and in fact f is strictly decreasing on $x > 0$.

The sketch

We need the behaviour at the two ends.

As $x \to 0^+$: $x/\sinh x \to 1$ and $(\cosh x)^{1/3} \to 1$, so $\mathrm{f}(x) \to 1$. In more detail, $(\cosh x)^{1/3} = 1 + \frac{1}{6}x^2 + O(x^4)$ and $x/\sinh x = 1 - \frac{1}{6}x^2 + O(x^4)$, so the x^2 terms cancel; one more term of each series gives

$$\mathrm{f}(x) = 1 - \tfrac{1}{45}x^4 + O(x^6).$$

So the curve leaves the (excluded) point $(0, 1)$ very flat, with zero gradient.

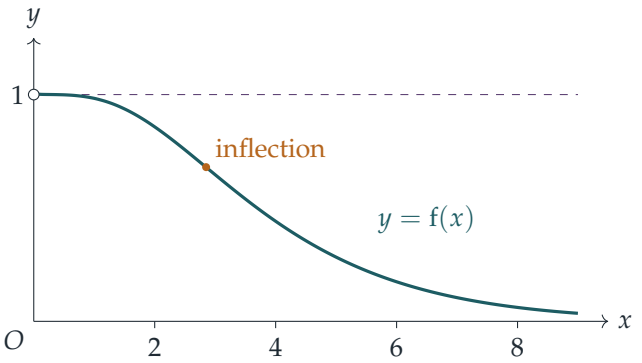
As $x \to \infty$: $\cosh x \sim \frac{1}{2}e^x$ and $\sinh x \sim \frac{1}{2}e^x$, so

$$\mathrm{f}(x) \sim \frac{x\,(\frac{1}{2}e^x)^{1/3}}{\frac{1}{2}e^x} = 2^{2/3}\,x\,e^{-2x/3} \to 0,$$

since the exponential beats the power of x. The x-axis is an asymptote, and the curve stays above it because $\mathrm{f} > 0$.

Ask yourself The curve starts flat and ends flat, yet always decreases. What does that force?

Near 0 the curve bends downwards (it is $1 - x^4/45$), and for large x it is a decaying exponential, which bends upwards. So there must be at least one point of inflection in between; numerically there is one, near $x \approx 2.85$, where $\mathrm{f} \approx 0.68$. Some values: $\mathrm{f}(1) \approx 0.983$, $\mathrm{f}(2) \approx 0.858$, $\mathrm{f}(4) \approx 0.441$, $\mathrm{f}(6) \approx 0.174$.



Checks

- At $x = 1$: $1 - \tanh 1 \approx 0.238$, $\sinh 1 - 2\cosh 1 + 2 \approx 0.089$ and $2\cosh 2 - 3\sinh 2 + 4 \approx 0.643$, all positive.
- The identity $\mathrm{f}'/\mathrm{f} = -k(x)/(6x\sinh x\cosh x)$ can be tested at any point: at $x = 1$ both sides are about -0.0592, and a numerical derivative of f agrees.
- Numerically $(1 - \mathrm{f}(x))/x^4 \approx 0.0222 \approx \frac{1}{45}$ for small x, and $\mathrm{f}(x)/(2^{2/3}xe^{-2x/3}) \to 1$ for large x.

Common slips

- Showing only that the derivative is positive, and forgetting to check that the function is 0 at $x = 0$; both are needed.
- Trying to differentiate f with the quotient and product rules directly. It works, but the logarithmic derivative is far cleaner and makes the link to (iii) visible.

- Drawing the curve crossing the x-axis, or starting at the origin. It starts at height 1 (not attained) and stays positive.

Beyond STEP

The technique of proving $F(x) > 0$ from $F(0) = 0$ and $F' > 0$ is the mean value theorem in action, and chaining it (as here, through k', h', g') is a standard way to prove inequalities between functions in analysis. The expansion $\mathrm{f}(x) = 1 - x^4/45 + \cdots$ is a taste of manipulating power series, which first-year analysis puts on a rigorous footing. A good place to read about both is Abbott, *Understanding Analysis*.

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